Oligopoly: Competition and Cooperation

Few sellers → strategic decisions → cartels → incentives to cheat → welfare and public policy

1. What makes a market an oligopoly

In perfect competition, a small firm can take the market price as given. In an oligopoly, a firm's production or pricing decision affects its competitors. The competitors may respond, so each firm must consider what the others are likely to do. This is the central economic difference.

DEFINITION Oligopoly and interdependence

An **oligopoly** is a market dominated by **a few important sellers**, offering identical or differentiated products. Entry can be difficult. **Interdependence** means that one firm's best decision depends on the actions of its rivals. **Game theory** studies decisions made under this kind of strategic interaction. A **duopoly** is an oligopoly with two sellers.

A few suppliers of the same industrial input may compete principally by choosing production quantities. Several firms selling differentiated software can compete in price, quality, service, and advertising. In both cases the behavior of one firm changes the incentives of the others.

Market structure	Sellers and products	Main economic feature
Perfect competition	Many; identical products	Each firm takes price as given
Monopolistic competition	Many; differentiated products	Each firm faces downward-sloping demand
Oligopoly	A few dominant firms; products may differ	Choices depend on competitors' responses
Monopoly	One seller without close substitutes	The firm faces the market demand curve

KEY RESULT Conditions students must recognize

An oligopoly has **few significant competitors**, often **barriers to entry**, and above all **strategic interdependence**. It can supply homogeneous goods, as in the cement model below, or differentiated goods. A market's precise classification depends on the economically relevant competitors, not just the number of brand names.

So what? A firm may earn more by producing less if its rival also limits output. But it may earn still more by quietly expanding while the rival holds back. Understanding that conflict is the purpose of this chapter.

2. Collusion and cartels

> **DEFINITION Collusion and cartel**
>
> **Collusion** occurs when competing firms coordinate prices or output rather than compete independently. A **cartel** is a group of sellers that acts together to restrict output or set prices, attempting to obtain monopoly-like profits. Explicit agreements between competitors to fix prices are illegal in the United States.

Worked example: The regional cement market

Northgate Cement and **Riverside Cement** are the region's only producers of identical cement. Both have constant marginal and average variable cost of \$20 per ton and no fixed costs. Quantity Q is measured in tons per week and includes both firms' production. Market demand is

$$\boxed{P = 200 - \frac{Q}{10}} \qquad \text{and} \qquad \boxed{\text{Firm profit} = (P - 20)q.}$$

Here P is dollars per ton and q is one firm's weekly output. Higher combined output means a lower price for every ton sold. The demand schedule is:

Total Q (tons/week)	Price P (per ton)
0	\$200
900	\$110
1,050	\$95
1,200	\$80
1,350	\$65
1,800	\$20

The competitive benchmark

If the firms behaved as price takers, production would expand until the market price equals marginal cost: $P = MC = \$20$. Demand then gives $Q = 1{,}800$. With no fixed costs, each ton sells for exactly its \$20 cost, so economic profit is zero. This is also the efficient quantity in this simple model: the value of the last ton equals its resource cost.

The cartel outcome and an equal split

If Northgate and Riverside coordinate as a single monopolist, they restrict total output to $Q = 900$. Price rises to \$110. Joint profit equals $(110 - 20) \times 900 = \$81{,}000$. With an equal production agreement, each supplies 450 tons and earns $90 \times 450 = \$40{,}500$. Restricting supply benefits the producers as a group, but reduces the amount available to buyers.

What happens if just one firm cheats?

Suppose Northgate raises output from 450 to 600 tons while Riverside sticks to 450. Total quantity becomes 1,050, so price falls to \$95. Northgate's profit becomes $(95 - 20) \times 600 = \$45{,}000$. Riverside receives only $(95 - 20) \times 450 = \$33{,}750$. Northgate therefore gains by breaking the agreement, even though the combined industry profit has declined.

What happens if both firms cheat?

If each produces 600 tons, total output becomes 1,200 and price falls to \$80. Each earns $(80 - 20) \times 600 = \$36{,}000$. Both now earn less than the \$40,500 that each would receive if the cartel held together.

Does either firm want to change from 600?

Hold Riverside's output at 600. If Northgate raises production to 750, total output becomes 1,350 and price drops to \$65: profit becomes $(65 - 20) \times 750 = \$33{,}750$. If Northgate instead cuts production to 450, total output becomes 1,050 and price rises to \$95: profit becomes $(95 - 20) \times 450 = \$33{,}750$. Both deviations yield less than \$36,000. Because the firms have identical costs and demand conditions, the same reasoning applies to Riverside. Indeed, with this straight-line demand and constant cost, intermediate changes cannot improve profit either. Thus 600 tons each is a stable best-response outcome.

Why cheating initially pays: two opposing effects

When Northgate expands from 450 to 600 while Riverside stays at 450, the extra 150 tons sell at a \$75 margin (price \$95 minus cost \$20), adding $150 \times 75 = \$11{,}250$. But the price falls \$15 on Northgate's original 450 tons, reducing profit on those sales by $450 \times 15 = \$6{,}750$. The net gain is \$4,500, equal to 45,000 – 40,500. The *output effect* favors expansion; the *price effect* works against it.

> **KEY RESULT Output and price across market structures**
>
> **Cartel/monopoly:** $Q = 900$, $P = \$110$. **Noncooperative duopoly:** $Q = 1{,}200$, $P = \$80$. **Competitive benchmark:** $Q = 1{,}800$, $P = \$20$.
>
> Quantity increases, and price falls, as competition replaces successful coordination.

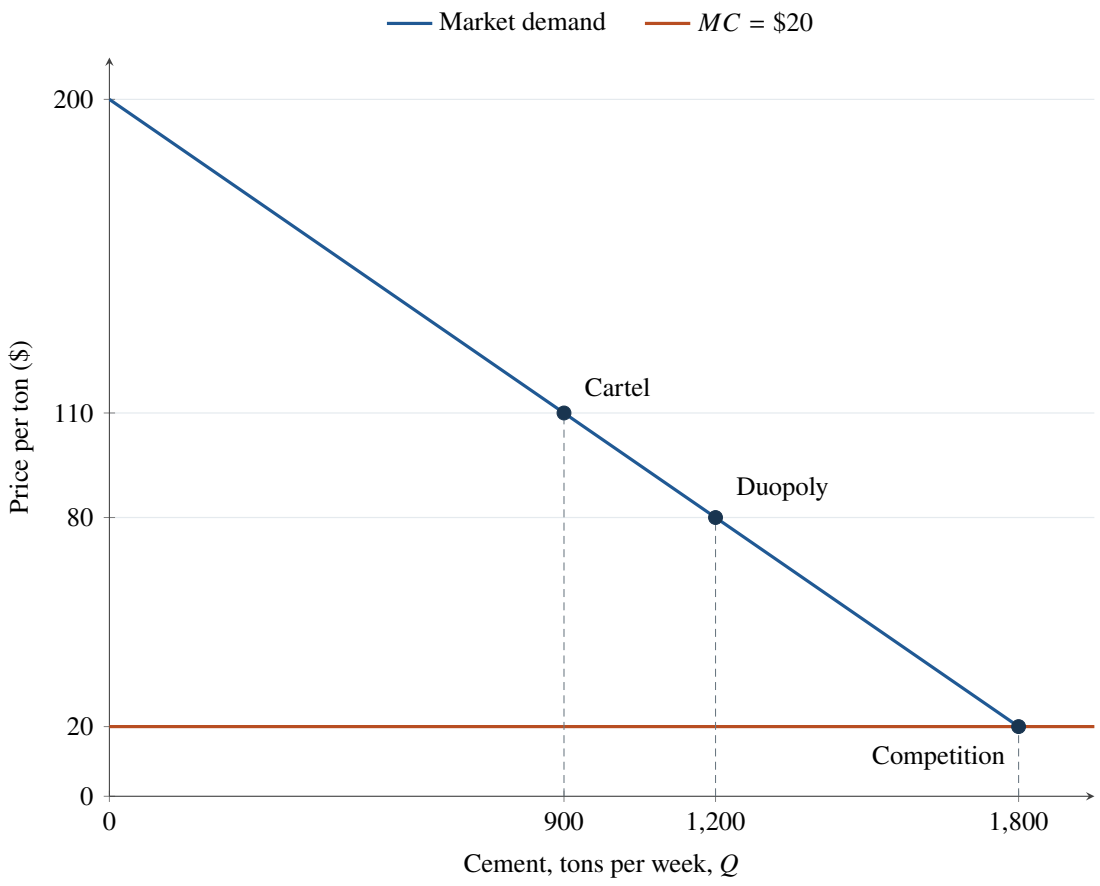


Figure 1. The same demand curve connects the three outcomes: 900 tons at $110 for the cartel, 1,200 tons at $80 for noncooperative duopoly, and 1,800 tons at $20 for the competitive benchmark.

Read the graph. Start at each marked quantity on the horizontal axis and move up to demand to read the price. More output pushes the price down. The competitive point is where demand meets marginal cost; the cartel restricts output well before that point.

So what? Each firm benefits from restricting *joint* output but has an incentive to produce more when it expects its rival to cooperate. That is why cartel agreements are fragile even before considering antitrust enforcement.

3. Why cartels break down: the prisoners' dilemma

A *payoff matrix* lists what each firm earns for each combination of decisions. Read the row as Northgate's choice, the column as Riverside's choice, and the first number in each cell as Northgate's profit. The second number is Riverside's profit. Compare one firm's choices while holding the other's decision fixed.

> **DEFINITION Dominant strategy, Nash equilibrium, prisoners' dilemma**
>
> A **dominant strategy** yields a better payoff regardless of what the other player does (among the strategies shown). A **Nash equilibrium** is a combination of strategies in which **no player can improve its payoff by changing its own choice alone**. A **prisoners' dilemma** occurs when individually attractive strategies leave both players worse off than mutual cooperation would.

	Riverside's output	
Northgate's output	**450 tons**	**600 tons**
450 tons	($40,500, $40,500)	($33,750, $45,000)
600 tons	($45,000, $33,750)	($36,000, $36,000)

Worked example: Find the dominant strategies without guessing

If Riverside makes 450 tons, Northgate earns $45,000 by making 600 rather than $40,500 by making 450. If Riverside makes 600, Northgate earns $36,000 by making 600 rather than $33,750 by making 450. So 600 is Northgate's dominant strategy *between the two choices in the table*. Riverside's incentives are identical. Both select 600, resulting in $36,000 each. The Nash equilibrium is worse for both than the cooperative $40,500 each.

The full quantity model strengthens this result: when Riverside makes 600, Northgate's maximum profit occurs at 600, not just among the two quantities in the table. A Nash equilibrium does not mean the firms achieve their highest *combined* profit. It means neither wants to deviate alone.

The same problem in advertising

Consider two other businesses that sell competing professional services. Their profits in this *separate* hypothetical game are measured in $ thousands (firm A first):

	Firm B's choice	
Firm A's choice	**No ad**	**Ad**
No ad	(50, 50)	(30, 60)
Ad	(60, 30)	(40, 40)

If B does not advertise, A prefers advertising: 60 exceeds 50. If B does advertise, A still prefers advertising: 40 exceeds 30. The same calculation applies to B. Both advertise and earn 40 each, even though both would receive 50 by not advertising. The problem is strategic incentives, not that either firm made an arithmetic mistake.

> **KEY RESULT A stable outcome need not maximize joint profit**
>
> In both games, a dominant strategy leads to a Nash equilibrium that gives each firm less profit than mutual restraint. A cooperative agreement is not self-enforcing merely because both would benefit from keeping it.

4. When cooperation survives: repeated play

The incentives can change if firms expect to compete again. When today's behavior changes tomorrow's response, a short-term gain from cheating may be outweighed by lost future profits. That possibility does not mean coordination is legal or guaranteed to succeed.

DEFINITION Repeated game and tit-for-tat

A **repeated game** is a strategic situation played over successive periods. Under **tit-for-tat**, a player starts by cooperating and then responds in the next period by doing what the rival did in the previous period. The possibility of retaliation may discipline current behavior.

Worked example: Three weeks of cement sales

If each producer supplies the cartel quantity of 450 tons in all 3 weeks, each earns $3 \times \$40{,}500 = \$121{,}500$. Suppose instead Northgate deviates in week 1, producing 600 while Riverside makes 450. It earns \$45,000 initially. If the deviation leads to both firms producing 600 in weeks 2 and 3, Northgate earns \$36,000 in each later week, or 45,000 + 36,000 + 36,000 = \$117,000 over the 3 weeks.

Deviating raises week-1 profit by \$4,500 but sacrifices \$4,500 relative to cooperation *in each subsequent week*. Northgate is \$4,500 worse off over the three weeks. The arithmetic illustrates the threat of future retaliation, not a proof that a known three-week game sustains collusion: choices in the final week can undermine the agreement.

So what? A firm deciding whether to cheat must consider how long customers and competitors will remain in the market, whether deviations can be observed, and whether retaliation is credible. Repeated interaction can make coordination more attractive, but cannot eliminate the incentive conflict in every setting.

5. Number of firms and welfare

The two-firm result lies between monopoly and perfect competition. More competitors generally make a firm's influence on market price smaller. The following outcomes use *the same regional cement demand and \$20 marginal cost* as the preceding sections. They are alternative numbers of identical, independently choosing firms, not observations from different markets.

Number of firms	Total Q (tons/week)	Price per ton
1	900	\$110
2	1,200	\$80
3	1,350	\$65
5	1,500	\$50
8	1,600	\$40
17	1,700	\$30

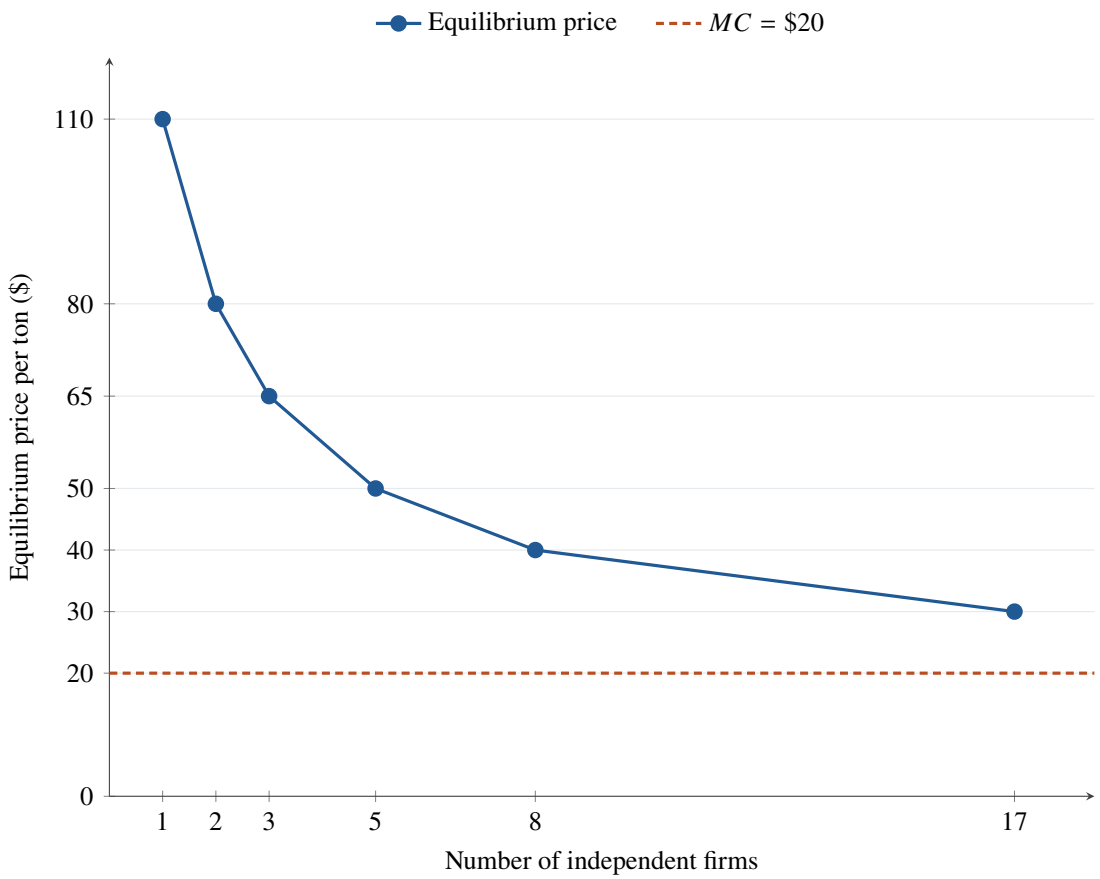


Figure 2. With the same demand and cost conditions, market price falls from $110 with 1 firm toward marginal cost of $20 as more sellers compete. Connecting segments guide the eye between the listed cases, not additional calculated equilibria.

Read the graph. Each dot corresponds to one row of the table. The downward path means greater competitive pressure, while the dashed $20 line is the limiting price benchmark. An oligopoly with 2 firms need not behave exactly like a monopoly, and 17 firms still need not be perfect competition.

What is the welfare loss from restricting production?

> **DEFINITION Consumer surplus and deadweight loss**
>
> **Consumer surplus** is buyers' total willingness to pay minus their actual spending. On a demand graph it is the triangle below demand and above price. **Deadweight loss** (DWL) is the value of beneficial trades that do not take place because output is restricted below the efficient quantity. It is the area between demand (marginal benefit) and marginal cost for the forgone units.

Worked example: Welfare from the same cement demand curve

Under competition, price equals \$20 and 1,800 tons are sold. Consumer surplus is $\frac{1}{2} \times 1{,}800 \times 180 = \$162{,}000$ and DWL is zero. With 2 noncooperating firms, price is \$80 and output is 1,200 tons; consumer surplus is $\frac{1}{2} \times 1{,}200 \times 120 = \$72{,}000$, while DWL is $\frac{1}{2} \times 600 \times 60 = \$18{,}000$. Under the cartel, price is \$110 and output only 900 tons; consumer surplus is $\frac{1}{2} \times 900 \times 90 = \$40{,}500$, and DWL is $\frac{1}{2} \times 900 \times 90 = \$40{,}500$.

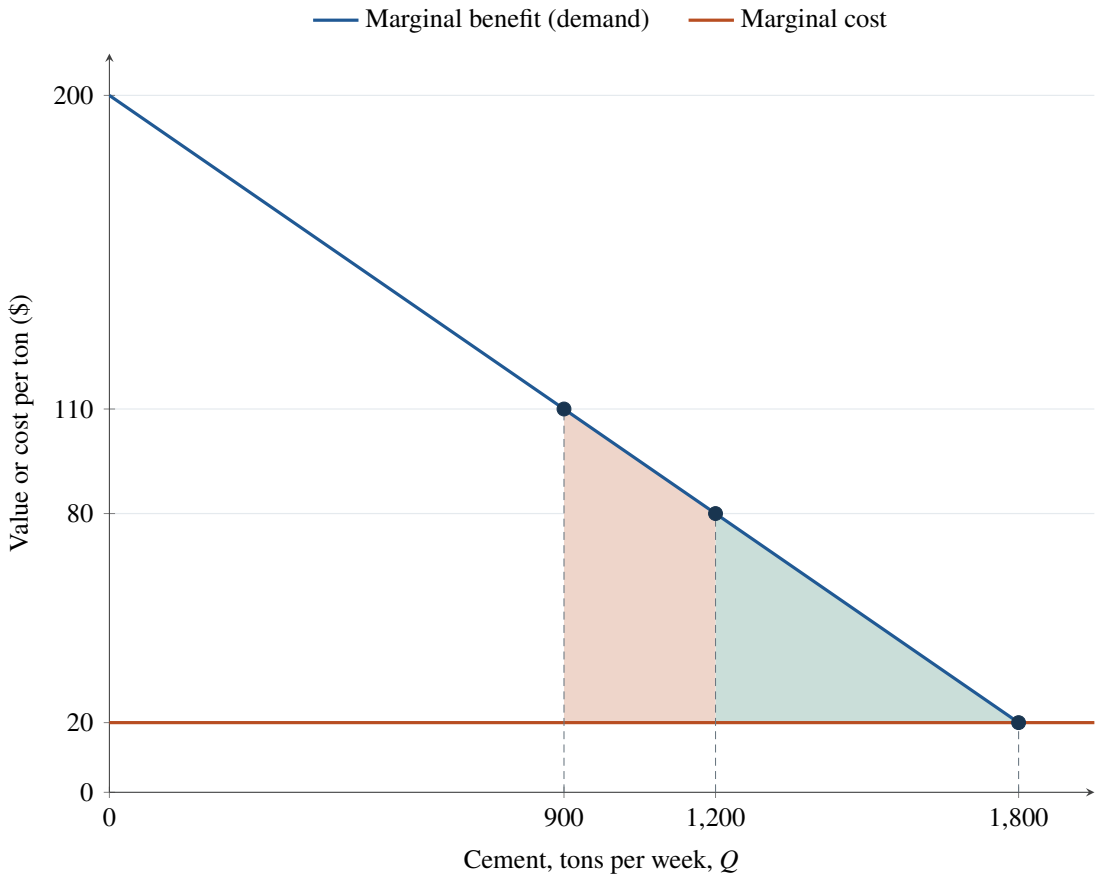


Figure 3. Green triangle: duopoly DWL of \$18,000 (output 1,200 to 1,800). Orange trapezoid: the additional \$22,500 lost when the cartel restricts output to 900. Together they are monopoly DWL of \$40,500. Both areas use the same demand and \$20 marginal cost.

Read the graph. All forgone transactions to the right of the 900-ton cartel quantity have willingness to pay above \$20 production cost. The area of those unrealized gains from trade is the entire shaded region. The noncooperative duopoly recovers the orange area by raising output to 1,200; the remaining green triangle disappears only at 1,800.

KEY RESULT Why buyers benefit when oligopolists compete

Collusion restricts output to raise price and moves surplus away from buyers. Independent competition raises quantity, lowers price, and reduces deadweight loss. Cartel profits include transfers from consumers, whereas **deadweight loss represents trades that never occur**, not merely money changing hands.

6. Public policy toward oligopolies

Because successful collusion can raise prices and restrict output, antitrust policy seeks to protect competition. But not every practice that limits an individual retailer's freedom or combines products is necessarily harmful; economists distinguish conduct that excludes competitors from conduct that reduces costs or improves service.

The principal U.S. antitrust laws

The Sherman Act (1890) prohibits agreements among competitors that unreasonably restrain trade, with price-fixing and market allocation treated as particularly serious violations. It also prohibits unlawful monopolization, allowing authorities to challenge exclusionary conduct rather than punishing size alone.

The Clayton Act (1914) addresses practices that may substantially lessen competition, including certain mergers and exclusive arrangements. It supplements the Sherman Act by reaching specific conduct before an industry has become fully monopolized.

Resale price maintenance

Resale price maintenance occurs when a manufacturer sets a minimum price at which independent dealers may resell its products. The *case against* it is that minimum resale prices can suppress competition among retailers and keep prices high. The *case for* it is that retailers providing demonstrations, technical advice, or after-sale service may lose sales to discount sellers whose customers use those services without paying for them. A minimum resale price can therefore protect useful services, although whether the practice is lawful or beneficial depends on its actual competitive effects.

Predatory pricing

Predatory pricing is a strategy of setting prices unusually low, potentially below an appropriate cost measure, to eliminate rivals and later charge higher prices. The *case against* it is the prospect of reduced competition and higher future prices. The *case for caution* is that low prices are also the main benefit of competition, and a firm that loses money to undercut rivals may be unable to recoup those losses if competitors return or new ones enter. A price reduction alone does not establish predation.

Bundling and tying

Bundling offers products together at a combined price; **tying** makes the purchase of one product conditional on purchasing another. The *case against* such arrangements is that a powerful supplier may foreclose rival suppliers of a linked product or weaken choice. The *case for* them is that combined sales can lower distribution costs, simplify purchasing, ensure compatibility, or help sellers price products to buyers with different valuations. Their effect on welfare cannot be inferred from the existence of a bundle alone.

So what? The competitive benchmark explains why authorities oppose cartels, but determining whether a particular pricing or product arrangement restricts competition requires examining the mechanism, evidence, and alternatives rather than assuming every unusual business practice is unlawful.

Quick review: the decisions and their consequences

Question	Economic answer
What defines oligopoly?	Few dominant sellers and strategic interdependence; entry may be difficult.
Why collude?	Restrict combined output to raise price and total profit toward the monopoly result.
Why is a cartel unstable?	A member can gain from additional sales while a rival keeps output restricted.
What is a Nash equilibrium?	No firm can earn more by changing its own action, holding its rival's action fixed.
What is a dominant strategy?	The best action whatever the rival does, among the available strategies.
Can repeated play help cooperation?	Future retaliation can outweigh the immediate benefit from deviating.
What happens as firms become more numerous?	Market quantity tends toward the efficient level; price tends toward marginal cost.
What is the welfare loss of collusion?	Missed gains from trade: output is below the efficient quantity.
What is antitrust policy for?	Prevent anticompetitive coordination and exclusion while weighing legitimate efficiencies.

Three common mistakes, corrected

Mistake 1: "If both firms would gain from a cartel, the cartel must survive." False. At 450 tons each, both earn $40,500, but a lone cheater producing 600 earns $45,000. Collective interest and individual incentives differ.

Mistake 2: "A Nash equilibrium is the outcome that maximizes total industry profit." False. The 600/600 outcome is stable at $36,000 each, even though both would prefer the cartel's $40,500 each. Stability is not the same as collective optimality.

Mistake 3: "Monopoly profit is itself the same as deadweight loss." False. Higher profit can reflect a transfer from buyers. Deadweight loss is the surplus destroyed by restricting production: $40,500 for the cartel, $18,000 for noncooperative duopoly, and zero at competitive output in this model.

> **KEY RESULT One final connection**
>
> **Oligopoly is a problem of strategic interdependence.** The same two firms can produce 900 tons through successful collusion or 1,200 tons when each chooses independently. Competing rather than coordinating raises output and lowers price. The fundamental tension is that a collective restriction of production can raise joint profit, while each individual firm may benefit from secretly increasing its own output.